

New high-end Dell display

Dell launches high-performance 24-inch LCD screen - Ultrasharp U2410

640 GB

Samsung spins out a 640 GB 2.5-inch hard drive

Motherboard test

One of the cheaper P45 boards, the Biostar TPowr I45 was one of the better performers of the lot. Intel's own DX48BT2 board was among the toppers in the class as well. The biggest variations were seen in the file transfer tests where all the hard drives were defragged before the tests and yet there were some differences. Some difference can be attributed to the difference in performance experienced because of using DDR3 or DDR2 memory. Game benchmarks such as *S.T.A.L.K.E.R Clear Sky* and *Crysis* were also almost identical throughout the test.

Intel Core 2 Boards (Integrated Graphics)

The Intel Core 2 processors can run on pretty much any chipset for LGA775. There aren't too many chipsets for Intel processors in recent times that provide decent graphics processing performance. The Intel G4x series of chipsets are pretty identical to the P4x series of chipsets, but they come with an onboard graphics solution. NVIDIA too has the GeForce 9300 range of chipset which we've already seen in the last comparison test. They matched the AMD 780G and also the NVIDIA GeForce 8200 and 8300 chipset in performance. We looked some of the newer Intel chipsets with onboard graphics - the G43 and the G41.

Features and design

These boards are really basic boards with the most basic of features. They are made for those who just want a desktop computer for everyday use and have no interest in running too many games. The Jetway I43GM1 is one of such boards and uses the Intel G43 chipset. The quality of heatsinks is cheap and so is the processor retention mechanism. There are only two memory slots which will let you install up to 8 GB of memory that might seem less as compared to some of the other motherboards. However, 8 GB still is a lot and it should

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be more than sufficient for most people. The board has an HDMI port which is a neat thing to have.

The MSI G43M2 is another G43 board, but this one comes with some really unexpected features such as the LPT and the COM ports. On the other extreme, there's a FireWire connector as well. The heat sink is slightly larger than the Jetway I43GM1-HG. The rest of the board sports standard features. There isn't anything special about this board but at the same time, there isn't anything wrong with it.

Like the Jetway I43GM1, the ECS G41T-M2 is another basic board and it looks like a really cheap one. That's the first impression you get. It looks like a stock Intel G41 board with its green PCB. The layout of the board is basic. The Southbridge has no cooling at all. The PCI slot is placed away but there are two PCI-E x1 slots in between but there is only just enough space for a card. The SATA ports are placed in the

doesn't use an all solid-state capacitor setup. There are only two RAM slots spanning most of the motherboard's length. This board is ideal for a really cheap setup but who want some quality. This kind of board suites those who want a DirectX 10 compatible board for use with Vista for example. Biostar like in many other board packages

Performance

Intel's onboard graphics solutions have been disregarded for the most part. NVIDIA's 9300 chipset motherboards have been around a while and you can read the reviews for those in our last motherboard test back in November last year. The cost of building systems around it has been a lot more than Intel chipsets. The Intel G43 and G41 chipset boards from Biostar, ECS, Jetway

and MSI didn't really break any records. We see that the G43 boards are just a tad faster than the G41 ones. The difference in real world performance is negligible and won't really produce any great frame rates even in pretty dated games. The boards sure match up with the other high-end boards even P45

in many of the video encoding and file transfer tests, so if you're building an Intel for an extremely low budget and just need a graphics solution built into the board, these chipsets will do just fine.

AMD Socket AM2+/AM3 (Discreet Graphics)

We talked about how most AMD chipsets have a decent graphics performance built into the chipset. The AMD 790FX and 790GX are almost identical chipsets. The 790GX runs a slightly more powerful version of the 780G graphics chip. Even then, boards of this kind are typically used by enthusiasts and those who go in for dedicated graphics cards. Similarly, NVIDIA has the 780a chipset which makes up the higher range which makes for great SLI-capable boards on the AMD platform.

MSI's K9N2 Diamond uses the NVIDIA nForce 780a chipset. MSI's heat sink design uses the usual spiralling heat pipes with thin heat-fins on them. They aren't very flimsy but they can be troublesome while attaching heatsinks like we learnt. The cooling solution works very well though. It's a little strange to see different kind of RAM clips used for half of the slots. Huge capacitors are used all across the board. A PCIe x1 X-Fi audio card is bundled along with the board like the high-end ASUS models. The power control buttons used aren't fancy like the other brands either. A good thing is the arrangement of all the SATA ports over the length of two SATA ports. The board package has almost everything you'll need. There are the extra back panels, SLI bridges of different lengths and power connector converters.

The Jetway HA07 Ultra belongs to the Hummer range of boards. As compared to Jetway's entry-level boards,



Jetway I43GM1-HG

way of the PCIe x1 slots so there might be some problems if you put in a long PCIe x1 card.

The Biostar G41D-M7 is also pretty similar to the ECS G41 board. This board too